

Chapter 4 : Analysis of Variance

One-Way Analysis of Variance (ANOVA) and multiple comparisons

```
In [3]: #import sys
        #{sys.executable} -m pip install statsmodels
```

Example

Automated hand sanitization plays a crucial role in promoting hygiene in public and private spaces.. In response to this, a company designed a portable hand sanitizer dispenser robot. Suppose the quality control engineer of the company wants to analyse the response time of four portable hand sanitizer dispenser robots before releasing them to the market..

1. What is the independent variable in this design?

Portable hand sanitizer dispenser robots

2. What is the dependent variable in this design?

Response time

3. Is there a significant difference in the mean response time of 24 tasks carried out by the four portable hand sanitizer dispenser robots?

```
In [29]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import statsmodels.api as sm
from statsmodels.formula.api import ols

# Define the data
Robot = (['Robot 1'] * 5 + ['Robot 2'] * 8 + ['Robot 3'] * 7 + ['Robot 4'] * 4)
Response_time = [
    6.33, 6.26, 6.31, 6.29, 6.40,
    6.26, 6.36, 6.23, 6.27, 6.19, 6.50, 6.19, 6.22,
    6.44, 6.38, 6.58, 6.54, 6.56, 6.34,
    6.58, 6.29, 6.23, 6.19, 6.21
]

# Create DataFrame
df = pd.DataFrame({'Robot': Robot, 'Response_time': Response_time})
```

```
# Display data
df
```

Out[29]:

	Robot	Response_time
0	Robot 1	6.33
1	Robot 1	6.26
2	Robot 1	6.31
3	Robot 1	6.29
4	Robot 1	6.40
5	Robot 2	6.26
6	Robot 2	6.36
7	Robot 2	6.23
8	Robot 2	6.27
9	Robot 2	6.19
10	Robot 2	6.50
11	Robot 2	6.19
12	Robot 2	6.22
13	Robot 3	6.44
14	Robot 3	6.38
15	Robot 3	6.58
16	Robot 3	6.54
17	Robot 3	6.56
18	Robot 3	6.34
19	Robot 3	6.58
20	Robot 4	6.29
21	Robot 4	6.23
22	Robot 4	6.19
23	Robot 4	6.21

4. Compute the mean and standard deviation for each robot

```
In [30]: # Group-wise summary
df.groupby('Robot').agg(
    count=('Response_time', 'count'),
    mean=('Response_time', 'mean'),
```

```
std=('Response_time', 'std')
)
```

Out[30]:

	count	mean	std
Robot			
Robot 1	5	6.318000	0.052631
Robot 2	8	6.277500	0.105255
Robot 3	7	6.488571	0.100570
Robot 4	4	6.230000	0.043205

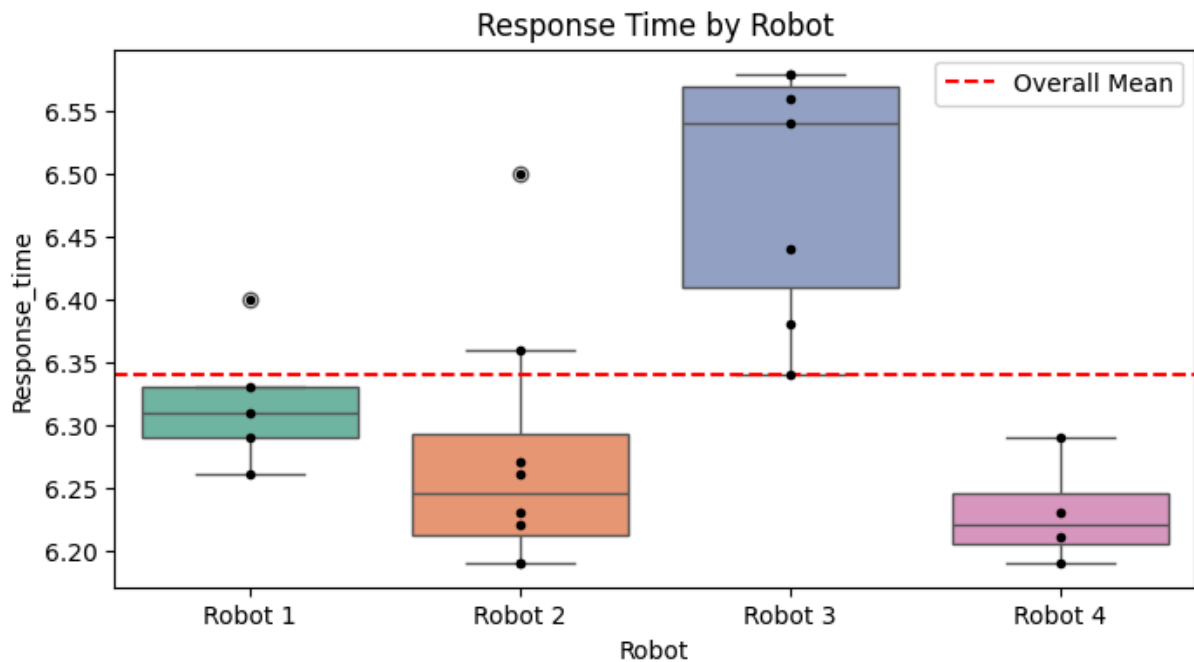
5. Graphically check if there is a difference between the distribution.

```
In [31]: plt.figure(figsize=(8, 4))

# Use 'Robot' for both x and hue, then remove the Legend
sns.boxplot(x='Robot', y='Response_time', hue='Robot', data=df, palette='Set2', dodg

# Overlay raw data points
sns.stripplot(x='Robot', y='Response_time', data=df, color='black', size=4, jitter=

# Add overall mean line
plt.axhline(df['Response_time'].mean(), color='red', linestyle='--', label='Overall
plt.legend()
plt.title('Response Time by Robot')
plt.show()
```



6. Use one-way ANOVA test to determine whether there is a significant difference in the mean response time of the four portable hand sanitizer dispenser robots.

Step 1: Establish null and alternative hypothesis

$$H_0: \mu_A = \mu_B = \mu_C = \mu_D$$

H_1 : At least one of the means is different from the others

Step 2: Determine the appropriate statistical test

one-way ANOVA

Step 3: Set the value of α , the Type I error rate.

$$\alpha = 0.05$$

Step 4: Establish the decision rule

If p-value is less than the significance level, we reject H_0 .

Step 5: Gather sample data

The observed data appear in the table.

Step 6: Analyse the data

```
In [42]: # Fit the model
model = ols('Response_time ~ C(Robot)', data=df).fit()

# Perform ANOVA
anova_table = sm.stats.anova_lm(model, typ=2)
anova_table
```

```
Out[42]:
```

	sum_sq	df	F	PR(>F)
C(Robot)	0.236580	3.0	10.181025	0.000279
Residual	0.154916	20.0	NaN	NaN

In the formula 'Response_time ~ C(Robot)', the C() function explicitly treats the variable Robot as a categorical (factor) variable, even if it's stored as numeric in your DataFrame.

The ANOVA table from statsmodels' anova_lm() does not include the Mean Square (MS) column by default

```
In [43]: # Add Mean Squares (MS) column
anova_table['MS'] = anova_table['sum_sq'] / anova_table['df']

# Reorder columns
anova_table = anova_table[['df', 'sum_sq', 'MS', 'F', 'PR(>F)']]

print(anova_table)
```

	df	sum_sq	MS	F	PR(>F)
C(Robot)	3.0	0.236580	0.078860	10.181025	0.000279
Residual	20.0	0.154916	0.007746	NaN	NaN

Step 7: Reach a statistical conclusion

From the above result, we can see that p-value is lower than the significance level (0.05). Therefore, we reject the null hypothesis.

Multiple comparison

- The one-way ANOVA test does not inform which robot (group) has a different mean.
- Multiple comparisons are to be used only when the analysis of variance yields an overall significant difference in the treatment means.

Tukey's test

- In analysis of variance, this technique is used for pairwise a posterior multiple comparisons to determine if there is a significant differences between the means of any pair of treatment levels in an experimental design.

```
In [35]: from statsmodels.stats.multicomp import pairwise_tukeyhsd

# Perform Tukey HSD
tukey = pairwise_tukeyhsd(df['Response_time'], df['Robot'])
print(tukey)
```

```
Multiple Comparison of Means - Tukey HSD, FWER=0.05
=====
group1 group2 meandiff p-adj lower upper reject
-----
Robot 1 Robot 2 -0.0405 0.8503 -0.1809 0.0999 False
Robot 1 Robot 3 0.1706 0.0169 0.0263 0.3148 True
Robot 1 Robot 4 -0.088 0.4613 -0.2532 0.0772 False
Robot 2 Robot 3 0.2111 0.0009 0.0836 0.3386 True
Robot 2 Robot 4 -0.0475 0.8144 -0.1983 0.1033 False
Robot 3 Robot 4 -0.2586 0.0008 -0.413 -0.1042 True
-----
```

Robot 3 is significantly slower than:

Robot 1 (meandiff = 0.1706, p = 0.0169)

Robot 2 (meandiff = 0.2111, p = 0.0009)

Robot 4 (meandiff = 0.2586, p = 0.0008)

- This clearly indicates Robot 3 has the worst performance in terms of response time.

Robots 1, 2, and 4 do not significantly differ from each other:

Robot 1 vs Robot 2: Not significant (p = 0.8503)

Robot 1 vs Robot 4: Not significant ($p = 0.4613$)

Robot 2 vs Robot 4: Not significant ($p = 0.8144$)

- These three robots have comparable and faster response times.

Test ANOVA assumptions

Assumption #1: Observations are drawn from normally distributed populations.

That is each sample is taken from a normally distributed population.

Assumption #2: Variances of the populations are equal.

That is the variance of data in the different groups should be the same.

Assumption #3: Observations represent random samples from the populations.

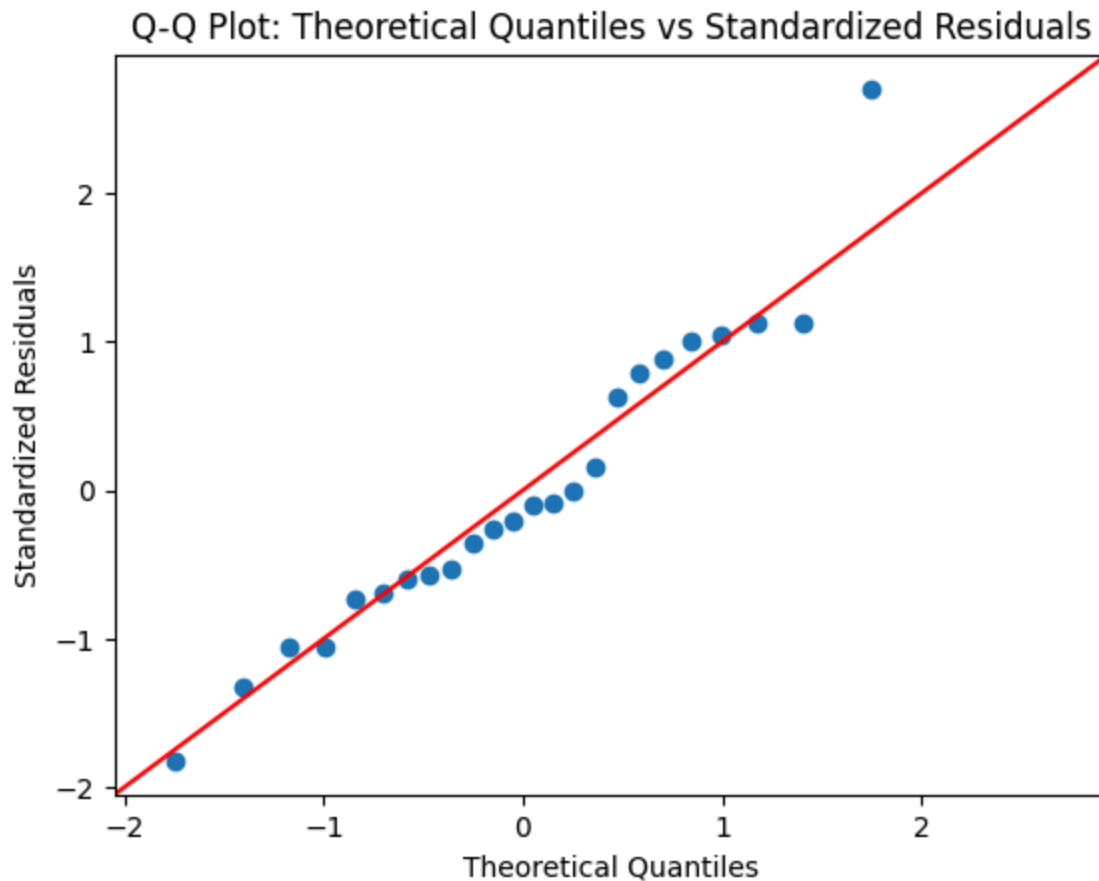
That is each sample has been drawn independently of the other samples.

Assumption 1: Normality of residuals

Plot: Q-Q Plot

If residuals follow a normal distribution, points will lie roughly on the 45-degree line.

```
In [48]: # Q-Q plot using standardized residuals
sm.qqplot(model.get_influence().resid_studentized_internal, line='45')
plt.title("Q-Q Plot: Theoretical Quantiles vs Standardized Residuals")
plt.xlabel("Theoretical Quantiles")
plt.ylabel("Standardized Residuals")
plt.show()
```



If the points mostly follow the straight line, the residuals are approximately normally distributed, satisfying the assumption. Deviations at the tails suggest mild skewness or kurtosis.

As the standardized residuals lie around the 45-degree line, it suggests that the residuals are approximately normally distributed

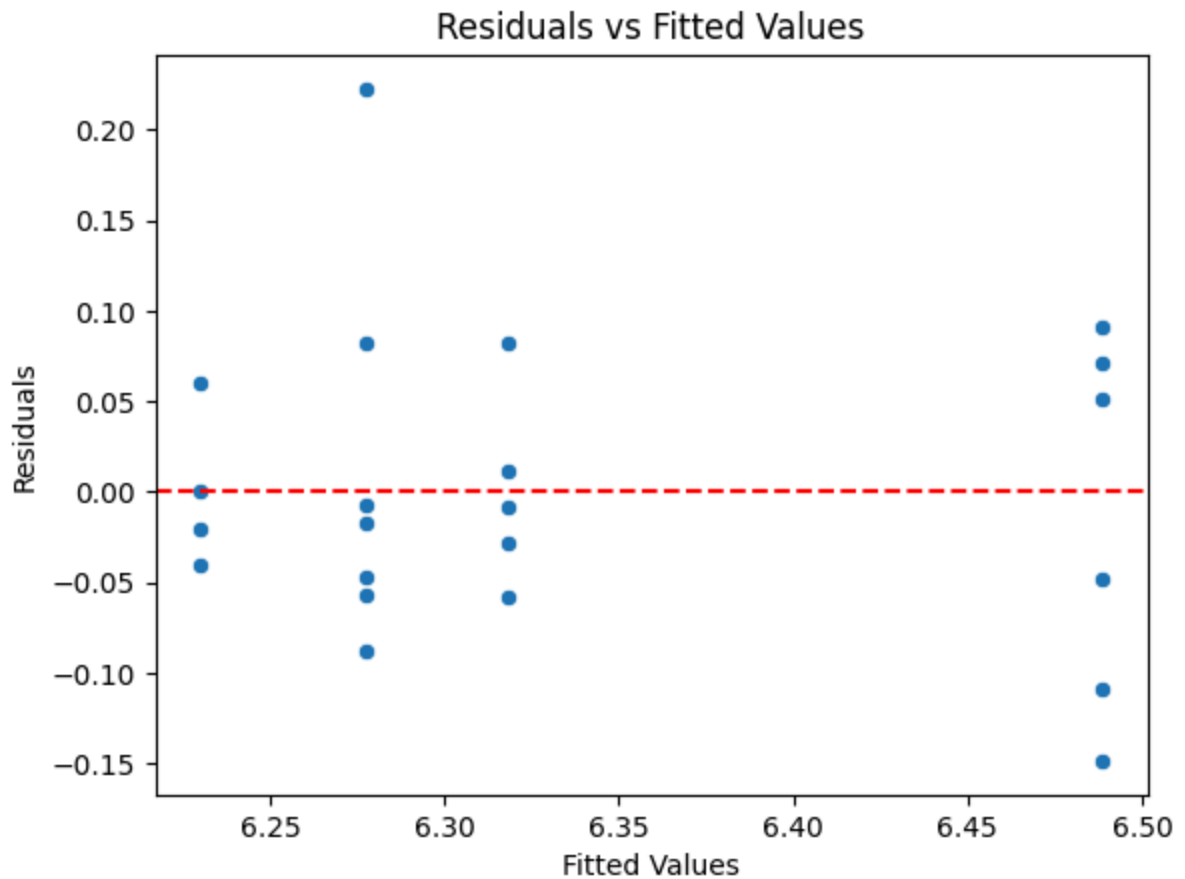
Assumption 2: Homogeneity of variances (equal variances)

Plot: Residuals vs Fitted Values

Statistical Test: Levene's Test (optional)

```
In [50]: # Create residuals vs fitted plot
fitted_vals = model.fittedvalues
residuals = model.resid

sns.scatterplot(x=fitted_vals, y=residuals)
plt.axhline(0, linestyle='--', color='red')
plt.xlabel("Fitted Values")
plt.ylabel("Residuals")
plt.title("Residuals vs Fitted Values")
plt.show()
```



The spread of residuals should be roughly constant across all fitted values (no funnel shape). If variance increases or decreases, the assumption is violated.

```
In [51]: # Levene's Test

from scipy.stats import levene

grouped = [group['Response_time'].values for name, group in df.groupby('Robot')]
stat, p_value = levene(*grouped)
print(f"Levene's test p-value: {p_value:.4f}")
```

Levene's test p-value: 0.5308

If $p > 0.05$, we fail to reject the null → Equal variances

If $p < 0.05$, we reject the null → Variances differ

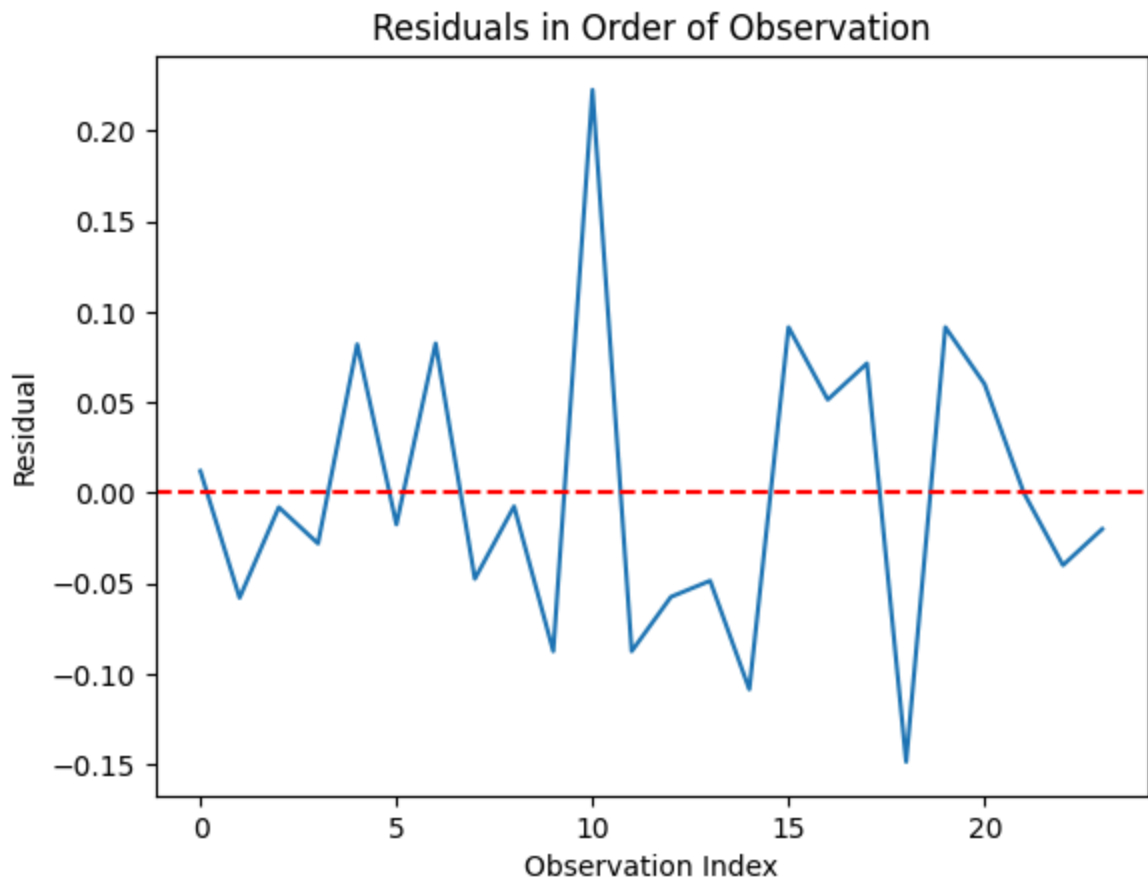
Assumption 3: Independence of observations

There's no standard plot for independence, but you can:

Ensure the data was collected in a randomized manner.

Residual plots should not show patterns (suggesting dependence).

```
In [52]: plt.plot(residuals)
plt.axhline(0, linestyle='--', color='red')
plt.title("Residuals in Order of Observation")
plt.xlabel("Observation Index")
plt.ylabel("Residual")
plt.show()
```



If the residuals fluctuate randomly around zero with no clear pattern, this supports the independence assumption.

All ANOVA assumptions were tested and reasonably satisfied. Therefore, the results and subsequent conclusions from the ANOVA and post-hoc tests are considered valid and reliable.

Step 8: Make a business decision

Robot 3 should not be released as-is. Its significantly higher response time (i.e., slower reaction) makes it less effective in practical use. Redesign or recalibration is recommended.

Robots 1, 2, and 4 demonstrate statistically similar and better performance. These can be approved for market release or considered for scaling production.

```
In [ ]:
```